

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
3 HT	(a)			Correct symbols for lamp and cell and variable resistor shown correctly in a series circuit (1) Accept variable power supply instead of cell and variable resistor If more than 1 cell shown they must be connected correctly in terms of polarity and no line passing through the cell Correct symbol for voltmeter and connected in parallel with their symbol for a lamp only (1) Correct symbol for ammeter and connected in series (1)	3			3		3
	(b)	(i)		Scale on x-axis of 1 V/cm and on y-axis of 0.1 A/cm (1) Accept scale on y-axis of 0.125 A/cm All points correctly plotted $\pm <1$ small square division – ignore (0,0) plot (1) Less than 5 points correctly plotted $\pm <1$ small square division – ignore (0,0) plot (0) Smooth curve between 0 – 12 V with $\pm <1$ small square division (1)		3		3	3	3